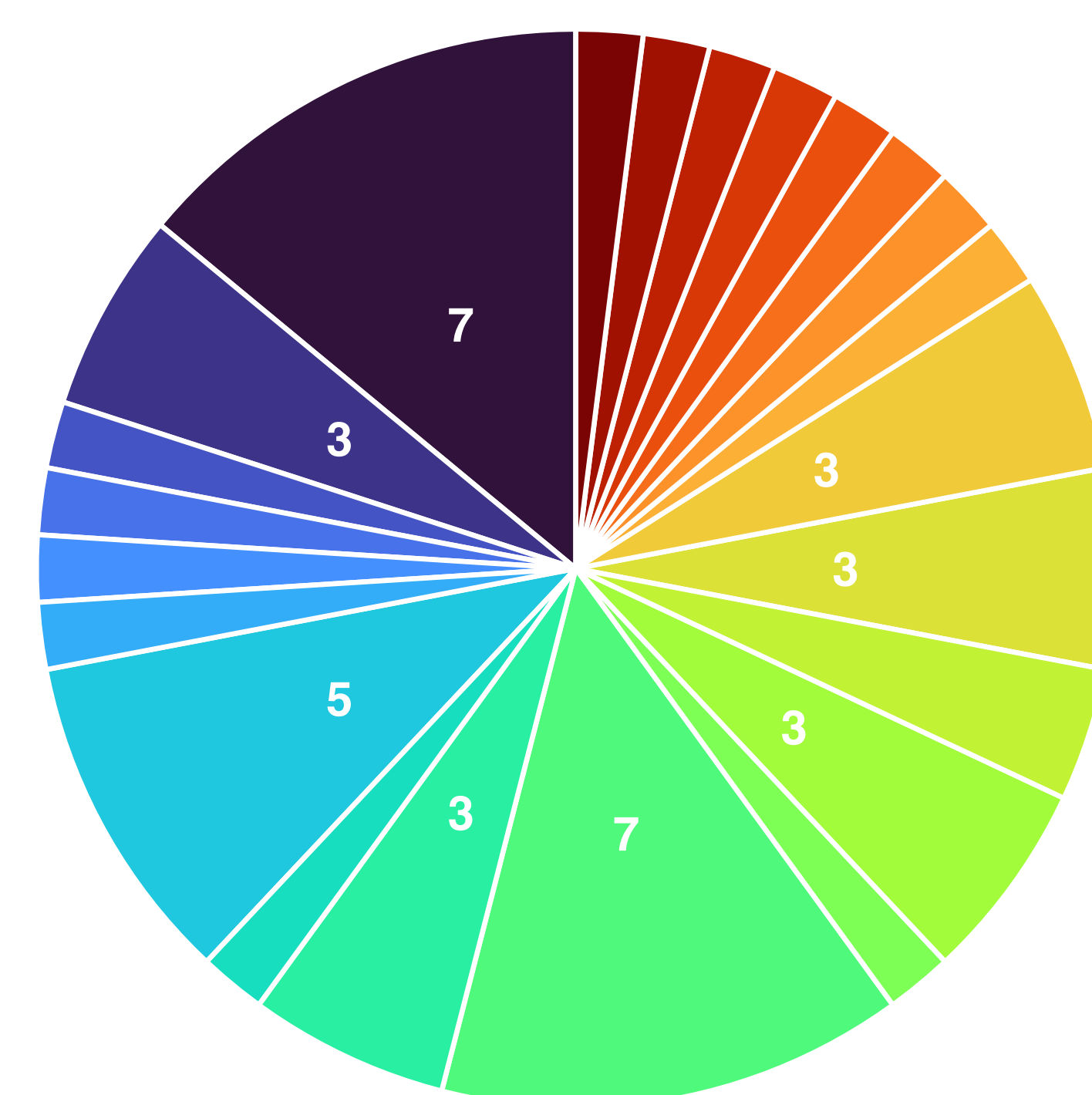
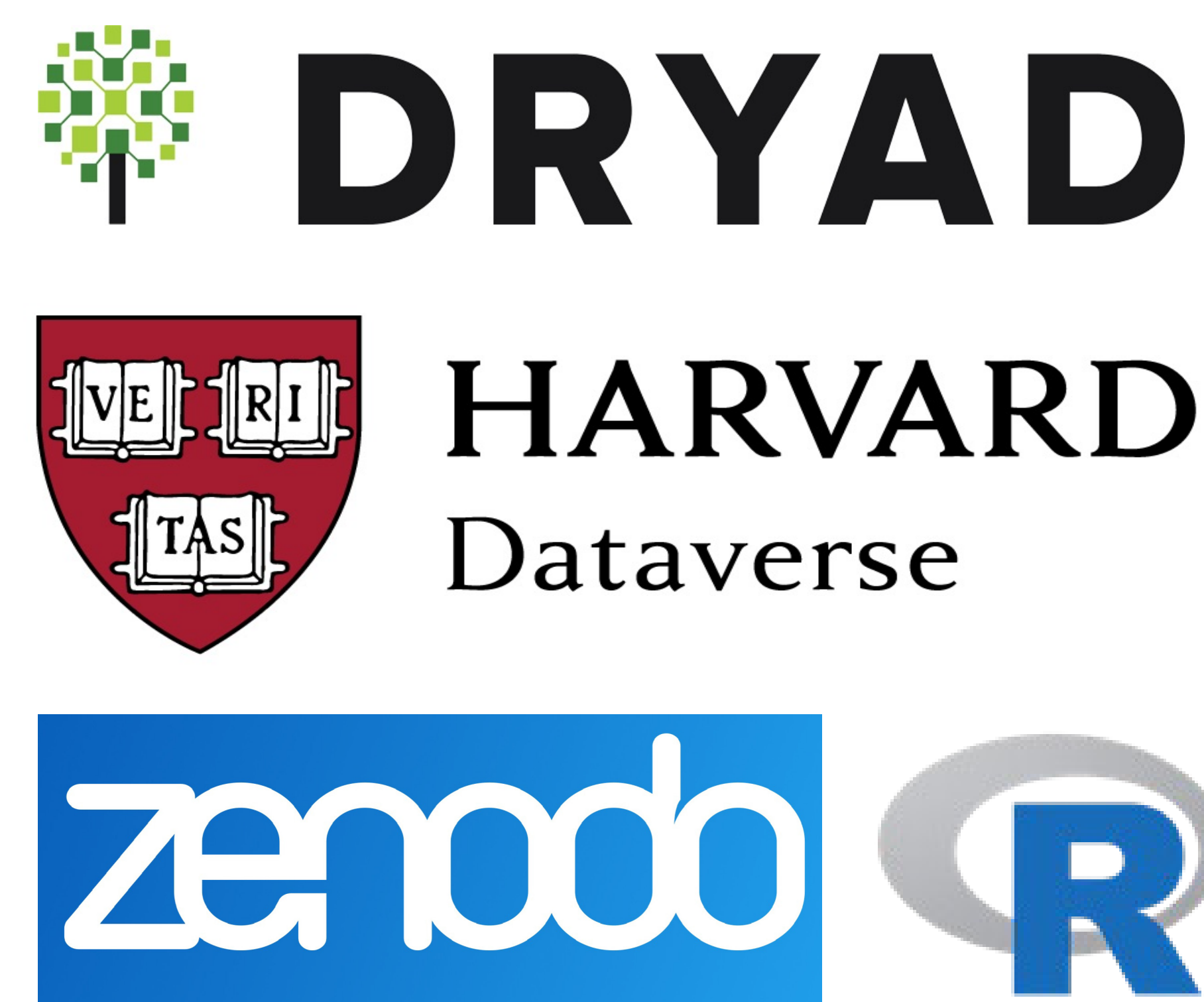


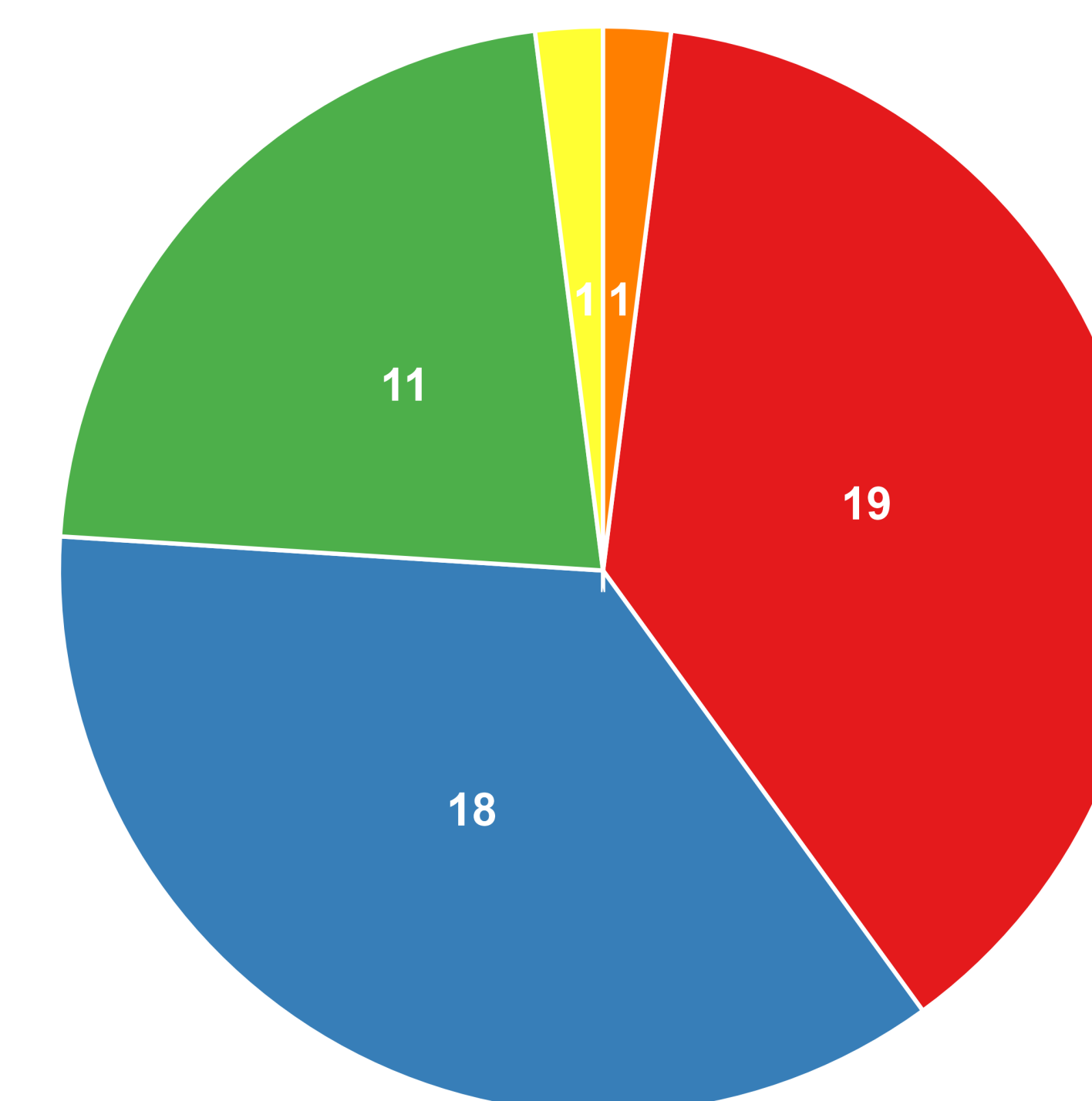
A

Data curation



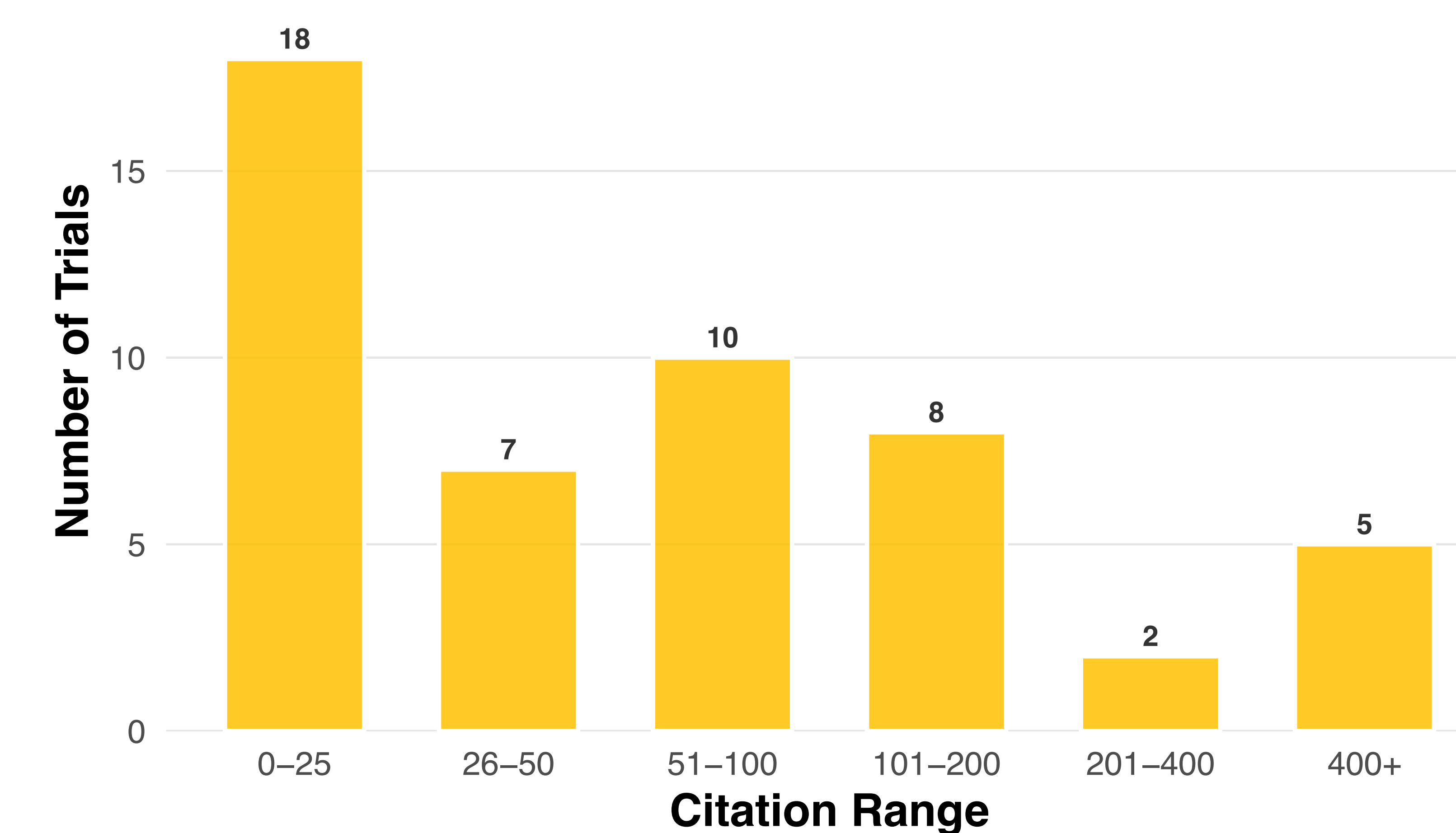
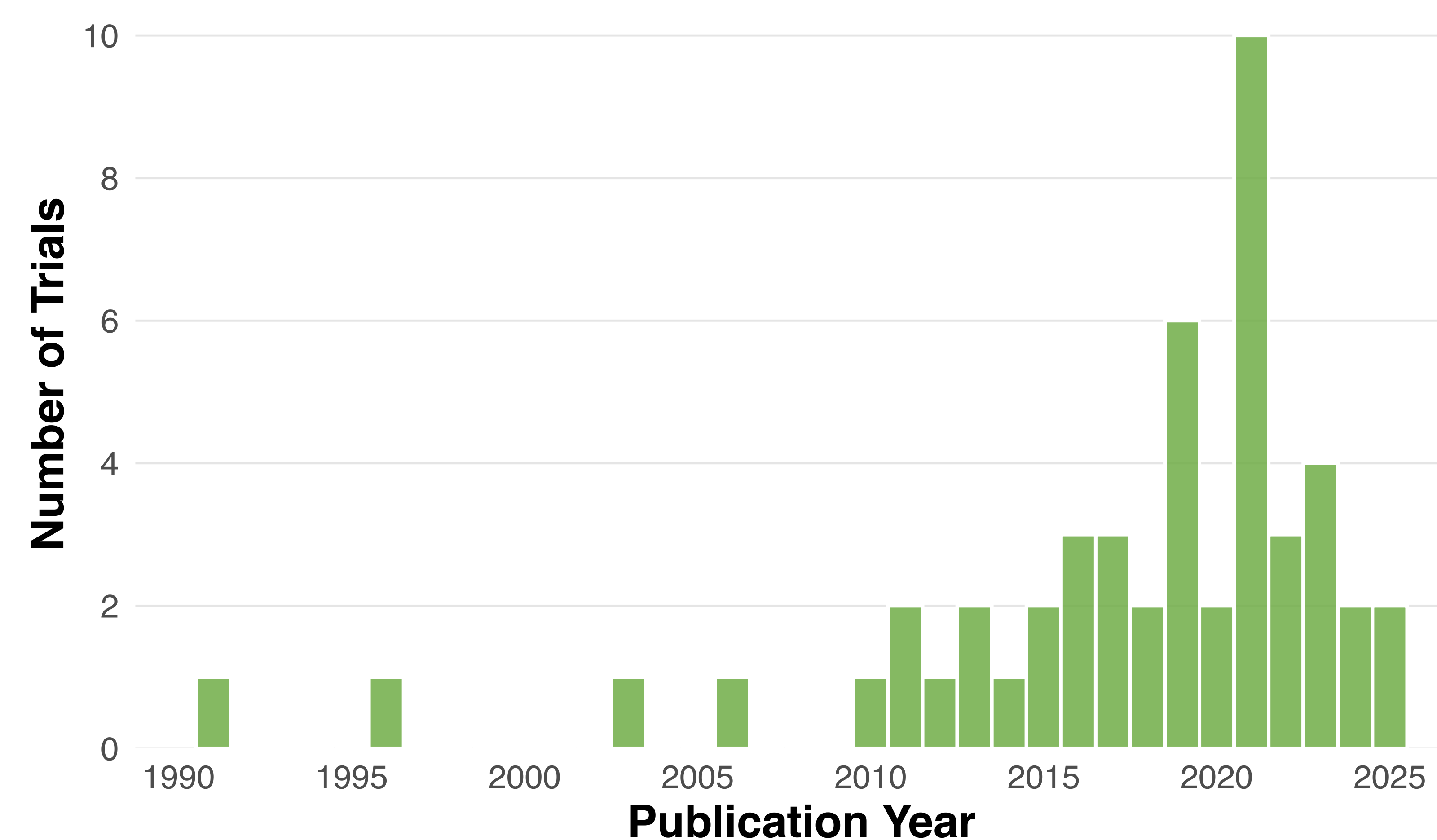
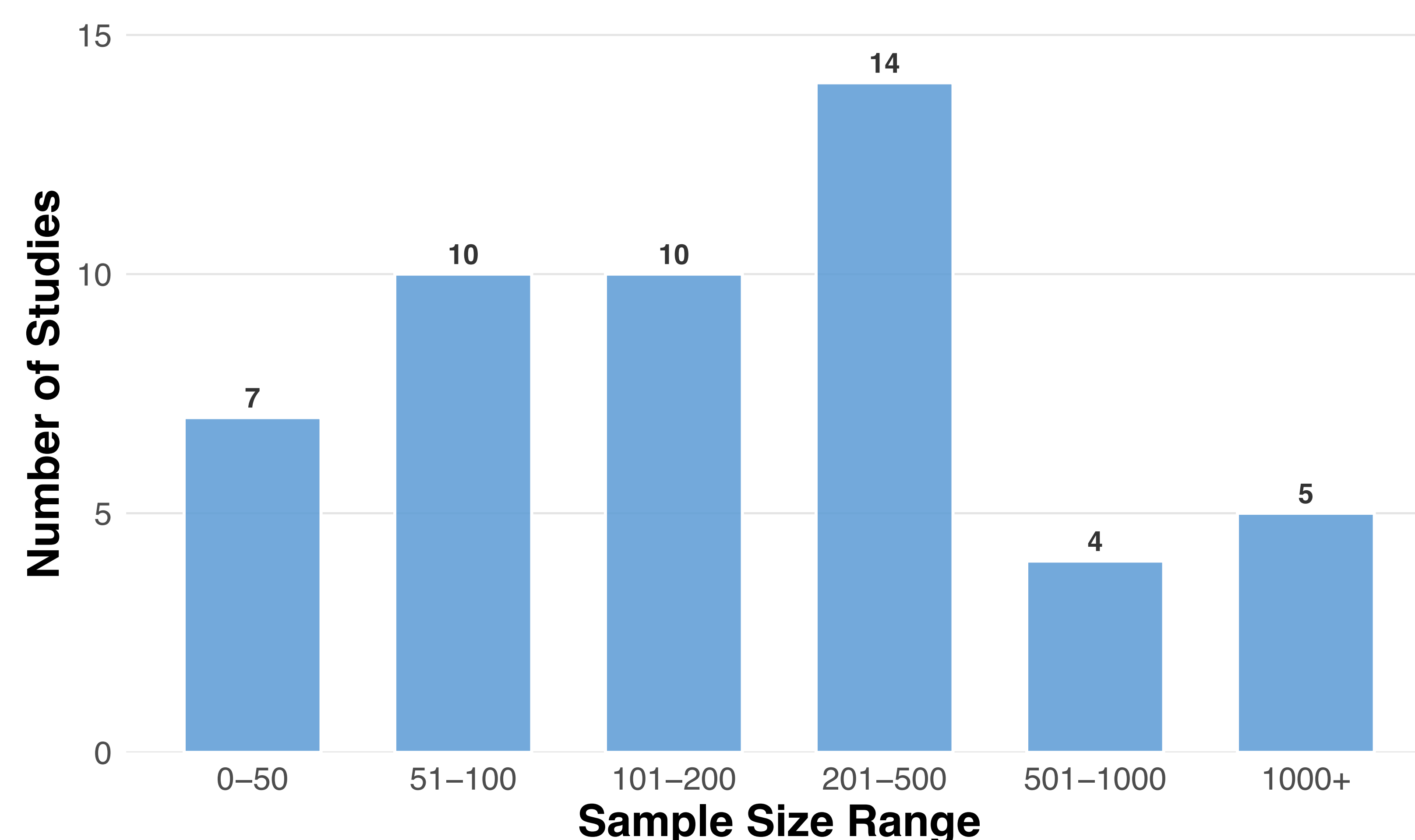
Research Area

Anesthesiology
Cardiovascular
Critical Care
Endocrinology
Gastroenterology
Hepatology
HIV/AIDS
Immunology
Infectious Disease
Maternal & Child Health
Medical Education
Mental Health
Nephrology
Neurology
Nutrition
Orthodontics
Pediatric Rehabilitation
Pediatrics
Pulmonology
Rehabilitation
Research Ethics
Surgery/Transplantation
Type 2 Diabetes



Randomization Scheme

Biased Coin
Block
Simple
Stratified
Stratified Rerandomization



B

Covariate adjustment methods

Statistical models

Regression: ANCOVA, ANHECOVA, IPW, g-logistic
Machine learning: TMLE, DML

X

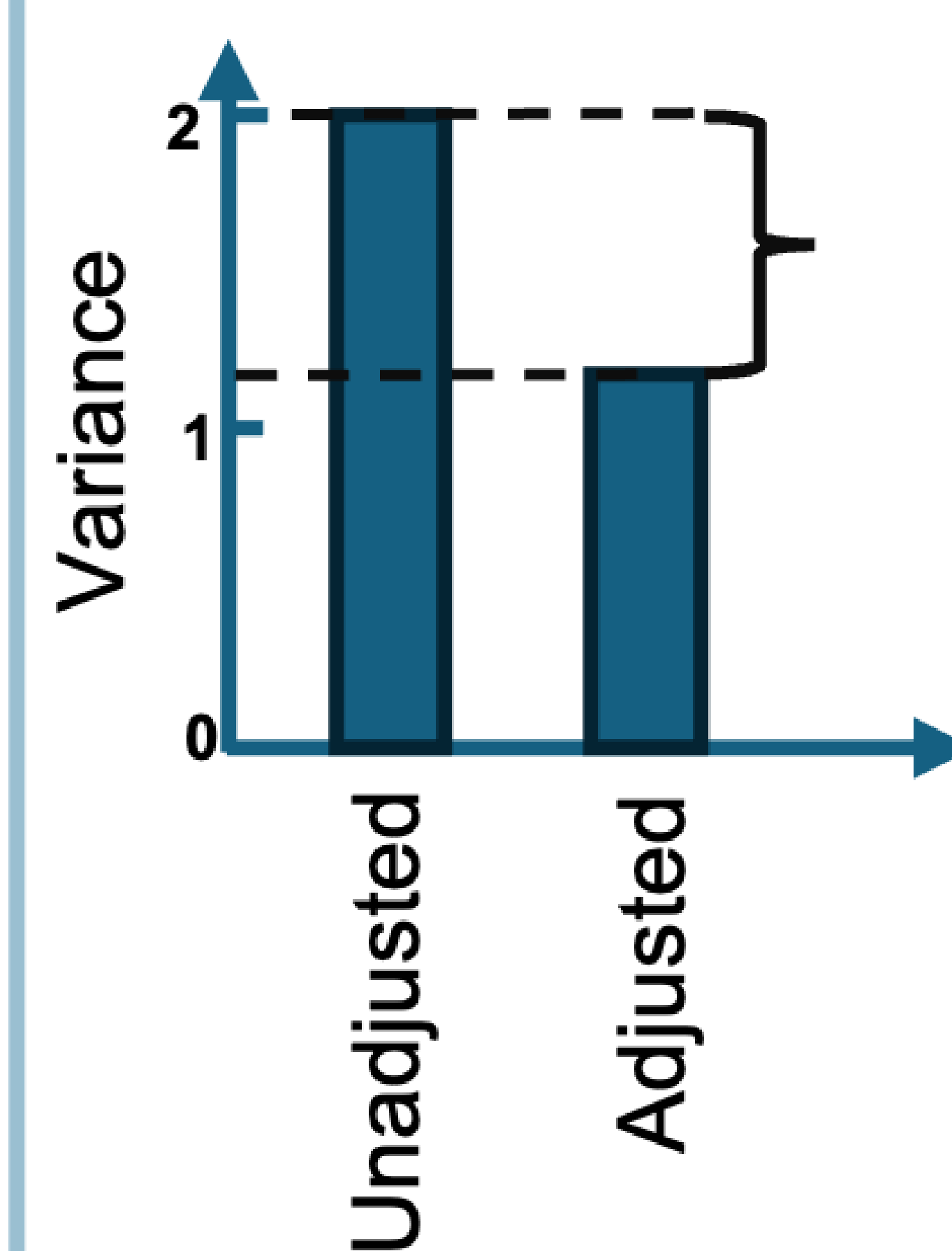
Covariate selection

- All
- Top-3
- Baseline+

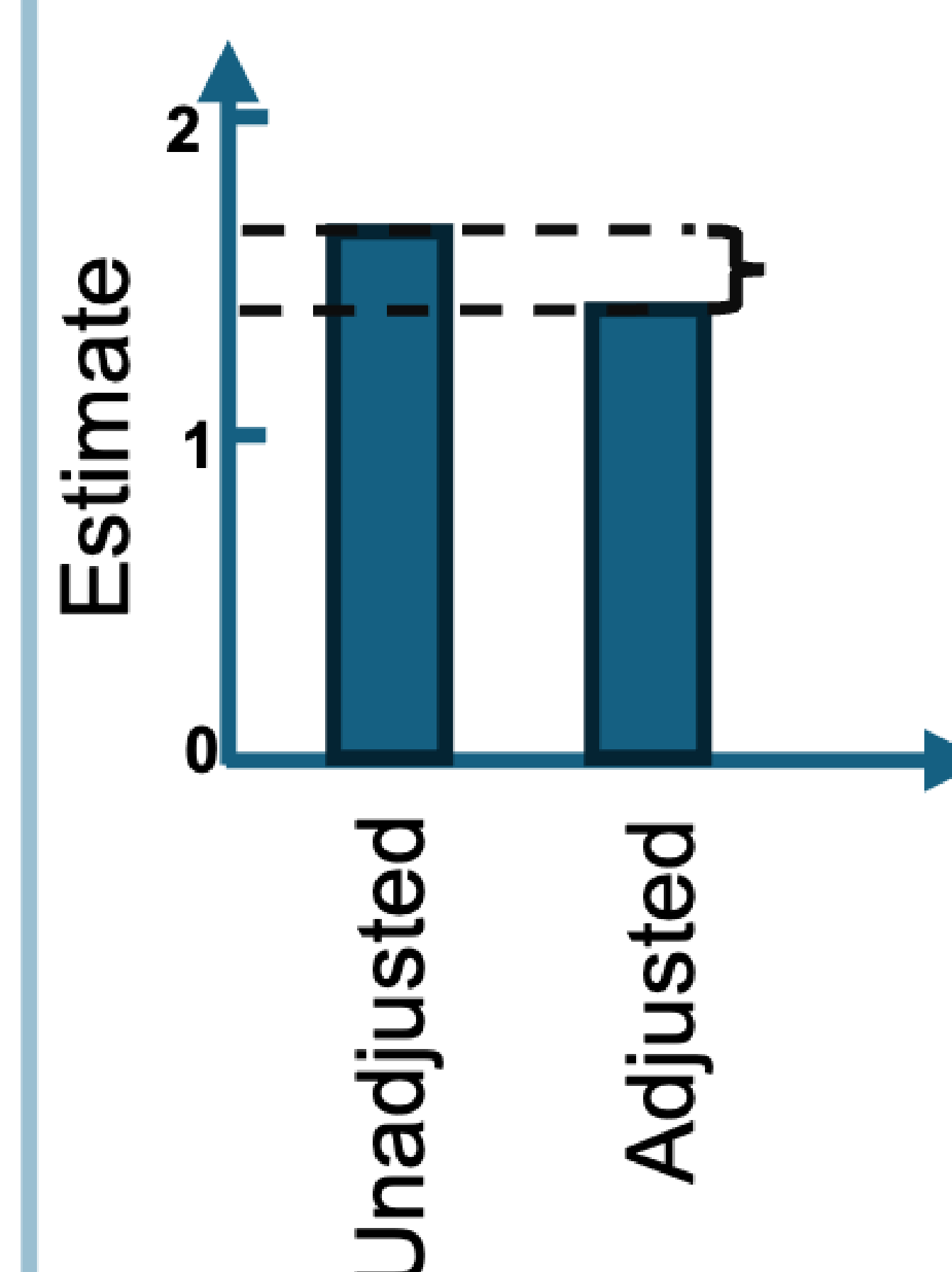
C

Performance metrics

Variance reduction



Estimate shift



Real-world impact

Unadjusted

Adjusted	Unadjusted	
	P > 0.05	P < 0.05
	P > 0.05	CAL
Error rate	P < 0.05	CAG